

AM-II Unit-II

Complex Analysis-II

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Contents

- 1 Laurent Series
- 2 Residue Integration Method
- 3 Residue Integration of Real Integrals
- 4 Geometry of Analytic Functions
- 5 Conformal Mapping
- 6 Linear Fractional Transformations (Mobius Transformations)
- 7 Special Linear Fractional Transformations
- 8 Conformal Mapping by Other Functions
- 9 Applications: Electrostatic Fields
- 10 Use of Conformal Mapping: Modeling
- 11 Heat Problems
- 12 Fluid Flow
- 13 Poisson's Integral Formula for Potentials

Laurent Series

Laurent series generalize Taylor series. If in application we want to develop a function $f(z)$ in powers of $z - z_0$ when $f(z)$ is singular at z_0 we can not use a Taylor series. Instead we may use a new kind of series, called Laurent series, consisting of positive powers of $z - z_0$ (and a constant) as well as negative integer powers of $z - z_0$.

Laurent's Theorem: Let $f(z)$ be analytic in a domain containing two concentric circles C_1 and C_2 with center z_0 and the annulus between them. Then $f(z)$ can be represented by the Laurent series.

$$\begin{aligned} f(z) &= \sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} \\ &= a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \dots + \frac{b_1}{z - z_0} \\ &\quad + \frac{b_2}{(z - z_0)^2} + \dots \end{aligned}$$

consisting of nonnegative and negative powers. The coefficients of this Laurent series are given by the integrals

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(z^*)}{(z^* - z_0)^{n+1}} dz^*, \quad b_n = \frac{1}{2\pi i} \oint_C (z^* - z_0)^{n+1} f(z^*) dz^*$$

taken counterclockwise around any simple closed path C that lies in the annulus and encircles the inner circle.

This series converges and represents $f(z)$ in the enlarged open annulus obtained from the given annulus by continuously increasing the outer circle C_1 and decreasing C_2 until each of the two circles reaches a point where $f(z)$ is singular.

In the important special case that z_0 is the only singular point of $f(z)$ inside C_2 , this circle can be shrunk to the point z_0 , giving convergence in a disk except at the center. In this case, the series of negative powers of the Laurent series is called the *principal part* of the singularity of $f(z)$ at z_0 .

Example

Find the Laurent series of $z^2 e^{\frac{1}{z}}$ with center 0.

Solution: We obtain a Laurent series whose principal part is an infinite series,

$$\begin{aligned} z^2 e^{\frac{1}{z}} &= z^2 \left(1 + \frac{1}{1!z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \frac{1}{4!z^4} + \dots \right) \\ &= z^2 + z + \frac{1}{2} + \frac{1}{6z} + \frac{1}{24z^2} + \dots \quad (|z| > 0). \end{aligned}$$

Example

Find the Laurent series of $z^{-5} \sin z$ with center 0.

Solution: We obtain

$$\begin{aligned} z^{-5} \sin z &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n-4} \\ &= \frac{1}{z^4} - \frac{1}{6z^2} + \frac{1}{120} - \frac{z^2}{5040} + \dots \quad (|z| > 0). \end{aligned}$$

Residue Integration Method

The purpose of Cauchy's integration method is the evaluation of integrals

$$\oint_C f(z) dz$$

taken around a simple close path C . The idea is as follows:

If $f(z)$ is analytic everywhere on C and inside C , such an integral is zero by Cauchy's integral theorem.

If $f(z)$ has a singularity at a point $z = z_0$ inside C , but is otherwise analytic on C and inside C , then $f(z)$ has a Laurent series

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \frac{b_1}{z - z_0} + \frac{b_2}{(z - z_0)^2} + \dots$$

That converges for all points near $z = z_0$ (except at $z = z_0$ itself), in some domain of the form $0 < |z - z_0| < R$.

The coefficient b_1 of the first negative power $\frac{1}{z-z_0}$ of this Laurent series is given by

$$b_1 = \frac{1}{2\pi i} \oint_C f(z) dz.$$

We obtain

$$\oint_C f(z) dz = 2\pi i b_1.$$

Here, we integrate counterclockwise around a simple closed path C that contains $z = z_0$ in its interior (but no other singular points of $f(z)$ on or inside C). The coefficient b_1 is called the residue of $f(z)$ at $z = z_0$ and we denote it by

$$b_1 = \text{Res}_{z=z_0} f(z).$$

Example

Integrate the function $f(z) = z^{-4} \sin z$ counterclockwise around the unit circle C .

Solution: We obtain the Laurent series

$$f(z) = \frac{\sin z}{z^4} = \frac{1}{z^3} - \frac{1}{3!z} + \frac{z}{5!} - \frac{z^3}{7!} + \dots$$

which converges for $|z| > 0$ (that is, for $z \neq 0$). This series shows that $f(z)$ has a pole of third order at $z = 0$ and the residue $b_1 = -\frac{1}{3!}$. Thus, we obtain

$$\oint_C \frac{\sin z}{z^4} dz = 2\pi i b_1 = -\frac{\pi i}{3}.$$

Residue Integration of Real Integrals

Consider integrals of the type

$$J = \int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta,$$

where $F(\cos \theta, \sin \theta)$ is a real rational function of $\cos \theta$ and $\sin \theta$ and is finite on the integration of integration. Setting $e^{i\theta} = z$, we obtain

$$\begin{aligned}\cos \theta &= \frac{1}{2}(e^{i\theta} + e^{-i\theta}) = \frac{1}{2}\left(z + \frac{1}{z}\right), \\ \sin \theta &= \frac{1}{2i}(e^{i\theta} - e^{-i\theta}) = \frac{1}{2i}\left(z - \frac{1}{z}\right).\end{aligned}$$

Since F is rational in $\cos \theta$ and $\sin \theta$, F is now a rational function of z , say, $f(z)$. As $d\theta = \frac{dz}{iz}$, therefore, the given integral takes the following form

$$J = \oint_C \frac{f(z)}{iz} dz.$$

As θ ranges from 0 to 2π , the variable z ranges counterclockwise once around the unit circle $|z| = 1$.

Example

Show that $\int_0^\infty \frac{dx}{1+x^4} = \frac{\pi}{2\sqrt{2}}$.

Solution: Indeed $f(z) = \frac{1}{1+z^4}$ has four simple poles at the points

$$z_1 = e^{\frac{\pi i}{4}}, \quad z_2 = e^{\frac{3\pi i}{4}}, \quad z_3 = e^{\frac{-3\pi i}{4}}, \quad z_4 = e^{\frac{-\pi i}{4}}.$$

The first two of these poles lie in the upper half-plane. The residues

$$\text{Res}_{z=z_1} f(z) = \left(\frac{1}{(1+z^4)'} \right)_{z=z_1} = \left(\frac{1}{4z^3} \right)_{z=z_1} = \frac{1}{4} e^{-\frac{3\pi i}{4}} = -\frac{1}{4} e^{\frac{\pi i}{4}},$$

$$\text{Res}_{z=z_2} f(z) = \left(\frac{1}{(1+z^4)'} \right)_{z=z_2} = \left(\frac{1}{4z^3} \right)_{z=z_2} = \frac{1}{4} e^{-\frac{9\pi i}{4}} = \frac{1}{4} e^{\frac{-\pi i}{4}}.$$

Therefore,

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^4} = -\frac{2\pi i}{4} (e^{\frac{\pi i}{4}} - e^{-\frac{\pi i}{4}}) = -\frac{2\pi i}{4} \sin \frac{\pi}{4} = \pi \sin \frac{\pi}{4} = \frac{\pi}{\sqrt{2}}$$

. Since $\frac{1}{1+x^4}$ is an even function, we thus obtain, as asserted,

$$\int_0^{\infty} \frac{dx}{1+x^4} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{dx}{1+x^4} = \frac{\pi}{2\sqrt{2}}.$$

Geometry of Analytic Functions

A complex function

$$w = f(z) = u(x, y) + iv(x, y) \quad (z = x + iy)$$

of a complex variable z gives a mapping of its domain D in the complex z -plane into the complex w -plane or onto its range of values in that plane. For any point z_0 in D the point $w_0 = f(z_0)$ is called the image of z_0 with respect to f . More generally, for the points of a curve C in D the image points form the image of C ; similarly for other point sets in D . Also, instead of the mapping by a function $w = f(z)$ we shall say more briefly the mapping $w = f(z)$.

Example

Consider the mapping $w = f(z) = z^2$. Using polar forms

$$z = re^{i\theta} \quad \text{and} \quad w = Re^{i\phi}$$

we have

$$w = z^2 = r^2 e^{2i\theta}.$$

Comparing moduli and arguments gives $R = r^2$ and $\phi = 2\theta$.

Hence, circles $r = r_0$ are mapped onto circles $R = r_0^2$ and rays $\theta = \theta_0$ onto the rays $\phi = 2\theta_0$. For the region $1 \leq |z| \leq \frac{3}{2}$, $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3}$, which is mapped onto the region $1 \leq |w| \leq \frac{9}{4}$, $\frac{\pi}{3} \leq \phi \leq \frac{2\pi}{3}$.

In Cartesian coordinates we have $z = x + iy$ and

$$u = \operatorname{Re}(z^2) = x^2 - y^2, \quad v = \operatorname{Im}(z^2) = 2xy.$$

Hence, vertical lines $x = c = \text{constant}$ are mapped onto $u = c^2 - y^2$, $v = 2cy$. From this we can eliminate y . We obtain $y^2 = c^2 - u$ and $v^2 = 4c^2y^2$. Together,

$$v^2 = 4c^2(c^2 - u).$$

These parabolas open to the left. Similarly, the horizontal lines $y = k = \text{constant}$ are mapped onto parabolas opening to the right

$$v^2 = 4k^2(k^2 + u).$$

A mapping $w = f(z)$ is called conformal if it preserves angles between oriented curves in magnitude as well as in sense. The angle $\alpha (0 \leq \alpha \leq \pi)$ between two intersecting curves C_1 and C_2 is defined to be the angle between their oriented tangents at the intersection point z_0 . And conformality, means that the images C_1^* and C_2^* of C_1 and C_2 make the same angle as the curves themselves in both magnitude and direction.

Conformality of mapping by analytic functions

The mapping $w = f(z)$ by an analytic function f is conformal, except at critical points, that is, points at which the derivative f' is zero.

Example

Conformality of $w = z^n$, $n = 2, 3, 4, \dots$

The mapping $w = z^n$, $n = 2, 3, 4, \dots$ is conformal except at $z = 0$, where $w' = nz^{n-1} = 0$. For general n , the angles at 0 are multiplied by a factor n under the mapping. Hence, the sector $0 \leq \theta \leq \frac{\pi}{n}$ is mapped by z^n onto the upper half-plane $v \geq 0$.

Example

Conformality of $w = z + \frac{1}{z}$.

The derivative of the mapping w is

$$w' = 1 - \frac{1}{z^2} = \frac{(z+1)(z-1)}{z^2}$$

which is 0 at $z = \pm 1$. These are the points at which the mapping is not conformal.

Linear Fractional Transformations (Möbius Transformations)

Conformal mappings can help in modeling and solving boundary value problems by first mapping regions conformally onto another. Linear fractional transformations (or Möbius transformations) are mappings

$$w = \frac{az + b}{cz + d} \quad (ad - bc \neq 0) \quad (1)$$

where a, b, c, d are complex or real numbers. Differentiation gives

$$w' = \frac{a(cz + d) - c(az + b)}{(cz + d)^2} = \frac{ad - bc}{(cz + d)^2}.$$

This motivates our requirement $ad - bc \neq 0$. It implies conformality for all z and excludes the totally uninteresting case $w' \equiv 0$ once and for all.

Special cases of (1) are

$$w = z + b \quad \text{(Translations)}$$

$$w = az \quad \text{with } |a| = 1 \quad \text{(Rotations)}$$

$$w = az + b \quad \text{(Linear transformations)}$$

$$w = \frac{1}{z} \quad \text{(Inversion in the unit circle).}$$

Example

Properties of the inversion $w = \frac{1}{z}$

In polar forms $z = re^{i\theta}$ and $w = Re^{i\phi}$ the inversion $w = \frac{1}{z}$ is

$$Re^{i\phi} = \frac{1}{re^{i\theta}} = \frac{1}{r}e^{-i\theta}$$

and gives $R = \frac{1}{r}$, $\phi = -\theta$. Hence, the unit circle $|z| = r = 1$ is mapped onto the unit circle $|w| = R = 1$; $w = e^{i\phi} = e^{-i\theta}$. For a general z , the image $w = \frac{1}{z}$ can be found geometrically by marking $|w| = R = \frac{1}{r}$ on the segment from 0 to z and then reflecting the mark in the real axis. We observe that $w = \frac{1}{z}$ maps horizontal and vertical straight lines onto circles or straight lines. Even the following is true:

$w = \frac{1}{z}$ maps every straight line or circle onto a circle or straight line.

Special Linear Fractional Transformations

Consider the linear fractional transformation

$$w = \frac{az + b}{cz + d} \quad (ad - bc \neq 0)$$

for mapping certain standard domains onto others.

Three points and their images given

Three given distinct points z_1, z_2, z_3 can always be mapped onto three prescribed distinct points w_1, w_2, w_3 by one, and only one, linear fractional transformation $w = f(z)$. This mapping is given implicitly by the equation

$$\frac{w - w_1}{w - w_3} \cdot \frac{w_2 - w_3}{w_2 - w_1} = \frac{z - z_1}{z - z_3} \cdot \frac{z_2 - z_3}{z_2 - z_1}.$$

(If one of these points is the point ∞ , the quotient of the two differences containing this point must be replaced by 1.)

Example

Find the linear fractional transformation that maps $z_1 = -1$, $z_2 = 0$, $z_3 = 1$ onto $w_1 = -1$, $w_2 = -i$, $w_3 = 1$, respectively.

Solution: We obtain

$$\frac{w - (-1)}{w - 1} \cdot \frac{-i - 1}{-1 - (-1)} = \frac{z - (-1)}{z - 1} \cdot \frac{0 - 1}{0 - (-1)}.$$

Thus,

$$w = \frac{z - i}{-iz + 1}.$$

Example

Determine the linear fractional transformation that maps $z_1 = 0$, $z_2 = 1$, $z_3 = \infty$ onto $w_1 = -1$, $w_2 = -i$, $w_3 = 1$, respectively.

Solution: We obtain

$$\frac{w - (-1)}{w - 1} \cdot \frac{-i - 1}{-1 - (-1)} = \frac{z - 0}{z - \infty} \cdot \frac{1 - \infty}{1 - 0}.$$

In this case, we replace the quotient $\frac{1-\infty}{z-\infty}$ by 1. Thus,

$$w = \frac{z - i}{z + i}.$$

Conformal Mapping by Other Function

Sine Function: Let

$$w = u + iv = \sin z = \sin x \cosh y + i \cos x \sinh y.$$

Therefore,

$$u = \sin x \cosh y \quad \text{and} \quad v = \cos x \sinh y.$$

Since $\sin z$ is periodic with period 2π , the mapping is certainly not one-to-one if we consider it in the full z -plane. We restrict z to the vertical strip $S : \frac{-\pi}{2} \leq x \leq \frac{\pi}{2}$. Since $f'(z) = \cos z = 0$ at $z = \pm \frac{\pi}{2}$, the mapping is not conformal at these two critical points. We claim that the rectangular net of straight lines $x = \text{constant}$ and $y = \text{constant}$ is mapped onto a net in the w -plane consisting of hyperbolas (the images of the vertical lines $x = \text{constant}$) and ellipses (the images of the horizontal lines $y = \text{constant}$) intersecting the hyperbolas at right angles.

As $\sin^2 x + \cos^2 x = 1$ and $\cosh^2 y - \sinh^2 y = 1$, we obtain

$$\frac{u^2}{\sin^2 x} - \frac{v^2}{\cos^2 x} = \cosh^2 y - \sinh^2 y = 1 \quad (\text{Hyperbolas}),$$

$$\frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = \sin^2 x + \cos^2 x = 1 \quad (\text{Ellipses}).$$

Exceptions are the vertical lines $x = \pm \frac{\pi}{2}$, which are folded onto $u \leq -1$ and $u \geq 1$ ($v = 0$), respectively.

The upper and lower sides of the rectangle are mapped onto semi-ellipses and the vertical sides onto $-\cosh 1 \leq u \leq -1$ and $1 \leq u \leq \cosh 1$ ($v = 0$), respectively. ($v = 0$), respectively.

Cosine Function: The mapping $w = \cos z$ could be discussed independently, but since

$$w = \cos z = \sin \left(z + \frac{\pi}{2} \right),$$

we see at once that this is the same mapping as $\sin z$ preceded by a translation to the right through $\frac{\pi}{2}$ units.

Hyperbolic Sine: Since

$$w = \sinh z = -i \sin(iz),$$

the mapping is a counterclockwise rotation $Z = iz$ through $\frac{\pi}{2}$ followed by the sine mapping $Z^* = \sin Z$, followed by a clockwise 90 degree rotation $w = -iZ^*$.

Hyperbolic Cosine: The function

$$w = \cosh z = \cos(iz)$$

defines a mapping that is a rotation $Z = iz$ followed by the mapping $w = \cos Z$.

Applications: Electrostatic Fields

The electrical force of attraction or repulsion between charged particles is governed by Coulomb's law. This force is the gradient of a function Φ , called the *electrostatic potential*. At any points free of charges, Φ is a solution of Laplace's equation

$$\nabla^2 \Phi = 0.$$

The surfaces $\Phi = \text{constant}$ are called *equipotential surfaces*. At each point P at which the gradient of Φ is not the zero vector, it is perpendicular to the surface $\Phi = \text{constant}$ through P , that is, the electrical force has the direction perpendicular to the equipotential surface.

The problems we shall discuss are two-dimensional, that is, we model physical systems that lie in three-dimensional space but are such that the potential Φ is independent of one of the space coordinates, so that Φ depends only on two coordinates, which we call x and y . Then Laplace's equation becomes

$$\nabla^2\Phi = \frac{\partial^2\Phi}{\partial^2x} + \frac{\partial^2\Phi}{\partial^2y} = 0.$$

Equipotential surfaces now appear as equipotential lines (curves) in the xy -plane.

Example

Find the potential Φ of the field between two parallel conducting plates extending to infinity which are kept at potentials Φ_1 and Φ_2 , respectively.

Solution: From the shape of the plates it follows that Φ depends only on x and Laplace's equation becomes $\Phi'' = 0$. By integrating twice we obtain $\Phi = ax + b$, where the constants a and b are determined by the given boundary values of Φ on the plates. For example. if the plates correspond to $x = -1$ and $x = 1$, the solution is

$$\Phi(x) = \frac{1}{2}(\Phi_1 - \Phi_2)x + \frac{1}{2}(\Phi_2 + \Phi_1).$$

The equipotential surfaces are parallel planes.

Use of Conformal Mapping. Modeling

A close relation results from the use of conformal mapping in modeling and solving boundary value problems for the Laplace equation, that is, in finding a solution of the equation in some domain assuming given values on the boundary ("Dirichlet problem"). Then conformal mapping is used to map a given domain onto one for which the solution is known or can be found more easily. This solution is then mapped back to the given domain. This is the idea. That it works is due to the fact that harmonic functions remain harmonic under conformal mapping.

Harmonic Functions Under Conformal Mapping

Let Φ^* be harmonic in a domain D^* in the w -plane. Suppose that $w = u + iv = f(z)$ is analytic in a domain D in the z -plane and maps D conformally onto D^* . Then the function

$$\Phi(x, y) = \Phi^*(u(x, y), v(x, y))$$

is harmonic in D .

Proof: The composite of analytic functions is analytic, as follows from the chain rule. Hence, taking a harmonic conjugate $\Psi^*(u, v)$ of Φ^* and forming the analytic function

$F^*(w) = \Phi^*(u, v) + i\Psi^*(u, v)$, we conclude that $F(z) = F^*(f(z))$ is analytic in D . Hence, its real part $\Phi(x, y) = \operatorname{Re} F(z)$ is harmonic in D .

Heat Problems

Laplace's equation also governs heat flow problems that are steady, that is, time-independent. Indeed, heat conduction in a body of homogeneous material is modeled by the *heat equation*

$$T_t = c^2 \nabla^2 T,$$

where the function T is temperature, $T_t = \partial T / \partial t$, t is time and c^2 is a positive constant (depending on the material of the body). Hence, if a problem is steady, so that $T_t = 0$ and two-dimensional, then the heat equation reduces to the two-dimensional Laplace equation

$$\nabla^2 T = T_{xx} + T_{yy} = 0,$$

so that the problem can be treated by our present methods.

$T(x, y)$ is called the *heat potential*. It is the real part of the *complex heat potential*

$$F(z) = T(x, y) + i\Psi(x, y).$$

The curves $T(x, y) = \text{constant}$ are called *isotherms* (= lines of constant temperature) and the curves $\Psi(x, y) = \text{constant}$ *heat flow lines*, because along them, heat flows from higher to lower temperatures.

The electrostatic equipotential lines $\Phi(x, y) = \text{constant}$ now become isotherms $T(x, y) = \text{constant}$, and the lines of electrical force become lines of heat flow.

Example

Find the temperature between two parallel plates $x = 0$ and $x = d$ having temperatures 0 and 100°C , respectively.

Solution: We see that $T(x, y) = ax + b$. From the boundary conditions, $b = 0$ and $a = 100/d$. The answer is

$$T(x, y) = \frac{100}{d} [^{\circ}\text{C}].$$

The corresponding complex potential is $F(z) = \frac{100}{d}z$. Heat flows horizontally in the negative x -direction along the lines $y = \text{constant}$.

Laplace's equation also plays a basic role in hydrodynamics, in steady nonviscous fluid flow under physical conditions. In order that methods of complex analysis can be applied, our problems will be two-dimensional, so that the velocity vector V by which the motion of the fluid can be given depends only on two space variables x and y and the motion is the same in all planes parallel to the xy -plane. Then we can use for the velocity vector V a complex function

$$V = V_1 + iV_2$$

giving the magnitude $|V|$ and direction $ArgV$ of the velocity at each point $z = x + iy$. Here V_1 and V_2 are the components of the velocity in the x and y directions. V is tangential to the path of the moving particles called a *streamline* of the motion.

We show that under suitable assumptions, for a given flow there exists an analytic function

$$F(z) = \Phi(x, y) + i\Psi(x, y),$$

called the *complex potential* of the flow such that the streamlines are given by $\Psi(x, y) = \text{constant}$ and the velocity vector or, briefly, the *velocity* is given by

$$V = V_1 + iV_2 = \overline{F'(z)},$$

where the bar denotes the complex conjugate. Ψ is called the *stream function*. The function Φ is called the *velocity potential*. The curves $\Phi(x, y) = \text{constant}$ are called *equipotential lines*. The velocity vector V is the *gradient* of Φ , by definition, this means that

$$V_1 = \frac{\partial \Phi}{\partial x}, \quad V_2 = \frac{\partial \Phi}{\partial y}.$$

Indeed, for $F = \Phi + i\Psi$, $F' = \Phi_x + i\Psi_x$ with $\Psi_x = -\Phi_y$ by the second Cauchy-Riemann equation.

Together we obtain

$$\overline{F'(z)} = \Phi_x - i\Psi_x = \Phi_x + i\Phi_y = V_1 + iv_2 = V.$$

Furthermore, since $F(z)$ is analytic, Φ and Ψ satisfy Laplace's equation

$$\begin{aligned}\nabla^2\Phi &= \frac{\partial^2\Phi}{\partial x^2} + \frac{\partial^2\Phi}{\partial y^2} = 0, \\ \nabla^2\Psi &= \frac{\partial^2\Psi}{\partial x^2} + \frac{\partial^2\Psi}{\partial y^2} = 0.\end{aligned}$$

Whereas in electrostatics the boundaries (conducting plates) are equipotential lines, in fluid flow the boundaries across which fluid cannot flow must be streamlines. Hence, in fluid flow the stream function is of particular importance.

Example

The complex potential $F(z) = z^2 = x^2 - y^2 + 2ixy$ models a flow with

Equipotential lines $\Phi = x^2 - y^2 = \text{constant}$ (*Hyperbolas*)
Streamlines $\Psi = 2xy = \text{constant}$ (*Hyperbolas*).

We obtain the velocity vector

$$V = 2\bar{z} = 2(x - iy),$$

that is, $V_1 = 2x$ and $V_2 = -2y$. The speed (magnitude of the velocity) is

$$|V| = \sqrt{V_1^2 + V_2^2} = 2\sqrt{x^2 + y^2}.$$

The flow may be interpreted as the flow in a channel bounded by the positive coordinates axes and a hyperbola, say, $xy = 1$. We note that the speed along a streamline S has a minimum at the point P where the cross section of the channel is large.

Poisson's Integral Formula for Potentials

Complex integration yields Poisson's integral formula for potentials in a standard domain (a circular disk). Poisson's formula will follow from Cauchy's integral formula

$$F(z) = \frac{1}{2\pi i} \oint_C \frac{F(z^*)}{z^* - z} dz^*. \quad (2)$$

Here C is the circle $z^* = Re^{i\alpha}$ (counterclockwise $0 \leq \alpha \leq 2\pi$) and we assume that $F(z^*)$ is analytic in a domain containing C and its full interior. Since $dz^* = iRe^{i\alpha} d\alpha = iz^* d\alpha$, we obtain

$$F(z) = \frac{1}{2\pi} \int_0^{2\pi} F(z^*) \frac{z^*}{z^* - z} d\alpha. \quad (3)$$

If instead of z inside C we take a Z outside C , the integrals (2) and (3) are zero by Cauchy's integral theorem.

We choose $Z = \frac{z^* \bar{z}^*}{\bar{z}} = \frac{R^2}{\bar{z}}$, which is outside C because $|Z| = \frac{R^2}{|z|} = \frac{R^2}{r} > R$. From (3) we thus have

$$0 = \frac{1}{2\pi} \int_0^{2\pi} F(z^*) \frac{z^*}{z^* - z} d\alpha = \frac{1}{2\pi} \int_0^{2\pi} F(z^*) \frac{z^*}{z^* - \frac{z^* \bar{z}^*}{\bar{z}}} d\alpha$$

and by straightforward simplification of the last expression on the right,

$$0 = \frac{1}{2\pi} \int_0^{2\pi} F(z^*) \frac{\bar{z}}{\bar{z} - \bar{z}^*} d\alpha.$$

We subtract this from (3) and use the following formula that you can verify by direct calculation ($\bar{z}z^*$ cancels):

$$\frac{z^*}{z^* - z} - \frac{\bar{z}}{\bar{z} - \bar{z}^*} = \frac{z^* \bar{z}^* - z \bar{z}}{(z^* - z)(\bar{z}^* - \bar{z})}.$$

We then have

$$F(z) = \frac{1}{2\pi} \int_0^{2\pi} F(z^*) \frac{z^* \bar{z}^* - z \bar{z}}{(z^* - z)(\bar{z}^* - \bar{z})} d\alpha.$$

From the polar representations of z and z^* we see that the quotient in the integrand is real and equal to

$$\frac{R^2 - r^2}{(Re^{i\alpha} - re^{i\theta})(Re^{-i\alpha} - re^{-i\theta})} = \frac{R^2 - r^2}{R^2 - 2Rr \cos(\theta - \alpha) + r^2}.$$

We now write $F(z) = \Phi(r, \theta) + i\Psi(r, \theta)$ and take the real part. Then we obtain *Poisson's integral formula*

$$\Phi(r, \theta) = \frac{1}{2\pi} \int_0^{2\pi} \Phi(R, \alpha) \frac{R^2 - r^2}{R^2 - 2Rr \cos(\theta - \alpha) + r^2} d\alpha.$$

This formula represents the harmonic function Φ in the disk $|z| \leq R$ in terms of its values $\Phi(R, \alpha)$ on the boundary (the circle $|z| = R$).

1. Advanced Engineering Mathematics by Erwin Kreyszig, John Wiley, 10th edition, 2011.